

BPM&EA Group Project and Test Plan

Submission: BPM&EA (UFCEP4-0-3)

Project: Hospital Patient Referral, Treatment and Administration System

Team: A/Ruby, B/Estrella, C/Ender, D/Ryan

Prepared: 29 September 2026

Shared implementation baseline reviewed: Git commit `fc6293e`

Executable process: `Hospital_All_Processes_Simple_C8`

1. Submission scope and shared artefacts

The BPM&EA and AISD submissions use the same case study and the same operational implementation. To avoid two divergent BPMN or Java copies, the authoritative executable model, eleven Camunda Forms, Java workers, Maven dependency file and configuration remain in `Coursework/hospital-pathway/`. This document is the BPM&EA-specific project and test record. AISD's separate acceptance and design-justification documents are in `submissions/AISD/`.

Operational BPMN and deployment configuration

Repository evidence: `Coursework/hospital-pathway/bpmn/W02_Hospital_All_Processes_Clean_Lines_Camunda8.bpmn`; deployment entry point `HospitalPathwayApplication.java`; `application.yaml`

Status at this review: Editable executable model and configuration present; one final-model self-pay route completed locally (G07).

External workers, dependencies and configuration template

Repository evidence: `Coursework/hospital-pathway/src/main/java/io/camunda/demo/hospital/`; `pom.xml`; `runnable src/main/resources/application.yaml`; `config/application.example.yaml`

Status at this review: Ten BPMN job types map to Java workers; a separate copyable configuration template is present; A/D automated checks pass; B/C behavioural checks remain open.

Editable Camunda Forms and task bindings

Repository evidence: `Coursework/hospital-pathway/forms/` (eleven `.form` files); `zeebe:formDefinition` on user tasks in the shared BPMN

Status at this review: Files and bindings deployed; G07 completed nine bound user tasks via REST with matching variables. Browser-side Tasklist form validation remains a separate check.

Project plan, test plan/results and evidence

Repository evidence: This document; `docs/p3-product-backlog.md`, `docs/p4-work-breakdown.md`, `docs/p6-sprint-backlog.md`, `docs/definition-of-done.md`; shared test logs in `Coursework/hospital-pathway/evidence/`

Status at this review: Planning sources and selected results present; gaps and failures retained below.

Repository and identifiable final version

Repository evidence: https://github.com/Estrella6066773/HW_AISD; implementation source baseline fc6293e; submission pack initially uploaded as e8a6861 on main

Status at this review: Repository upload completed; group second-owner review and any subsequent release tag remain open.

Presentation slides

Repository evidence: [submissions/BPMEA/Hospital_Pathway_BPMEA_Presentation_2026-09-29.pptx](#)

Status at this review: Eight editable BPM&EA slides cover the operational model, architecture, ownership, planning, recorded results, demo route and remaining criteria.

The older W01 assessment notes additionally call for a strategic BPMN, i* SD/SR and alignment evaluation for BPM&EA. The repository has a requirements traceability matrix ([docs/requirements-traceability-matrix.md](#)), but no clearly named final strategic BPMN or i* SD/SR deliverables were identified in this review. The group should check the tutor's definitive BPM&EA brief and attach those files if still required. This report does not relabel the operational Camunda model as a strategic or i* model.

2. Project planning and ownership

The prioritised product backlog ([docs/p3-product-backlog.md](#)) lists PB items with priority, requirement, estimate, first/second owner, dependency, acceptance condition, evidence and status. [docs/p4-work-breakdown.md](#) decomposes the work. The Sprint 1 backlog ([docs/p6-sprint-backlog.md](#)) targeted a plan increment and the first referral-to-booking slice. Its 16 September statuses are historical planning data, not a current completion claim. The Definition of Done requires an accessible artefact, review and evidence before an item is marked Done. Est/est: [PB-17贡献记录表.md](#) tracks individual contributions, with unreviewed items left open.

A / Ruby

First-owner process contribution: P5 clinical care, P7 funding, P8 urgent care, P13 reporting; finance exception routes

Current technical evidence: [RefundWorkers.java](#); five [RefundWorkersTest](#) results; S1-S10 group acceptance-plan draft

Handover/remaining evidence: Final-version refund, urgent-care and report demonstrations; second-owner sign-off.

B / Estrella

First-owner process contribution: P3/P4 visit and P6 treatment booking; shared H2 domain database

Current technical evidence: [BookingWorkers.java](#), [domain/entities/repositories](#), domain design note and earlier engine-route records

Handover/remaining evidence: Test pending/confirmed occupancy and retry in the final model; resolve source-level status transition mismatch.

C / Ender

First-owner process contribution: P1 registration, P2 referral/redirect and P9 clinic letter

Current technical evidence: `LetterWorkers.java`; BPMN job bindings; C implementation note

Handover/remaining evidence: Record live or automated letter/referrer receipt and retry results; second-owner sign-off.

D / Ryan

First-owner process contribution: P10 enquiries, P11/P12 care change

Current technical evidence: `CareChangeWorkers.java`; seven Java tests; four BPMN structural checks; historical PB-14/PB-18 REST runs

Handover/remaining evidence: Final-version care-change and enquiry runs; second-owner sign-off.

The hand-offs follow the shared BPMN rather than four isolated models: C's accepted referral reaches B's visit, A's clinical/funding work follows, and D's change route may return to A/Finance. Shared request-payment publishes the payment-result message, while send-booking-confirmation completes the booking communication step. The external payment, scheduling and correspondence services are classroom simulations.

3. Test strategy, evidence and verdict rules

Acceptance tests are specified against business rules and an identified process version. **PASS** means every recorded check in that scenario met its stated rule; **FAIL** means a check did not; **NOT EVIDENCED** means no matching execution record exists. A Java unit/integration test, static BPMN check, source file or earlier-version run is identified as such and is not substituted for final-version end-to-end acceptance. Use fictional case IDs only.

Existing evidence: `Ryan/2026-09-26/PB-14_acceptance-run-log_2026-09-29.json` and `PB-18_acceptance-run-log_2026-09-29.json` are local Camunda REST runs on an earlier deployed definition version 6. `Est/est: 全院流程路径记录.md` describes version 1 and `Est/est: 治疗预约与付款五条路径记录.md` a separate earlier booking/payment model. `Coursework/hospital-pathway/evidence/Java_Test_Results_2026-09-29.txt` records twelve passing A/D Java tests at `fc6293e`; `BPMN_Structural_Test_2026-09-29.txt` records four passing static checks. The new `Group_Final_E2E_20260929-135729.json` records a complete live Camunda REST run of the exact final BPMN source hash, locally deployed as definition version 1. Ruby's S1-S10 PDF is a plan, not a result log. The AISD detailed acceptance record gives individual data, actions and observations for G01-G11 and U01-U04.

Referral acceptance and rejection (BR-02/03)

Preconditions, data and actions: Version-6 model; unique case; submit Register then consultant Review as accept or reject.

Expected result / pass-fail rule: Accept opens Book visit and can reach Clinical care; reject ends with no booking. Fail if either route reverses.

Actual status and evidence: PASS for these slices on version 6 (PB-14 log); accepted instance stopped at Clinical care, so no complete-path claim.

Role boundary (BR-37)

Preconditions, data and actions: Version-6 Tasklist demo; submit a secretary-labelled `staffRole` with `accept`.

Expected result / pass-fail rule: Unauthorised acceptance refused and no Book visit.

Actual status and evidence: FAIL: Book visit was created (PB-18 AT-03). A role field does not enforce access.

Pending visit and treatment resource (BR-10/11)

Preconditions, data and actions: Accepted referral or authorised treatment; choose pending, then later ready for one case/window; inspect Tasklist plus `booking_slot`.

Expected result / pass-fail rule: Pending has no premature confirmation; a later ready result yields one CONFIRMED occupancy row. Fail on early confirmation, stale PENDING or duplicate row.

Actual status and evidence: Version-6 visit routing passed only a limited check. No current B database test. Source inspection identifies both a pending-visit CONFIRMED write risk and failure to promote an existing PENDING treatment row.

Failed payment and deliberate retry (BR-14/17)

Preconditions, data and actions: Funding path; mock amount 10.00, then human resolves issue and explicitly retries with 10.01.

Expected result / pass-fail rule: Failure waits at Funding issue and does not book treatment; deliberate retry can progress. No automatic second charge.

Actual status and evidence: PASS for recorded routing on version 6 (PB-18 AT-05); no real provider-charge proof.

Missing payment confirmation (BR-18)

Preconditions, data and actions: Funding path; choose investigation outcome.

Expected result / pass-fail rule: `payment_status=investigate`, human Funding issue opens, no automatic Book treatment.

Actual status and evidence: PASS for version-6 route (PB-18 AT-06); A's isolated worker checks pass.

Full self-pay path to end (BR-02/03/14/16/30)

Preconditions, data and actions: Final BPMN/forms and Java running; unique `GROUP-E2E-20260929-135729`; accept referral, attend, approve treatment, send letter, self-pay 10.01, reserve treatment and discharge.

Expected result / pass-fail rule: Correlated `payment-result`, required workers, no incident, final instance COMPLETED. Fail if stalled, wrong end or missing worker/message.

Actual status and evidence: PASS on one locally deployed final-model route: definition version 1, instance 2251799813685458, nine tasks, COMPLETED, zero active incidents, ten true assertions (G07 JSON). Tasks were completed via REST, so browser form validation and direct H2 row counts remain open. No successful `payment_ledger` row is claimed.

Letter/referrer notification (BR-30/31/32)

Preconditions, data and actions: Final model; approved letter and a separate redirected case; inspect variables and `audit_event`, retry each worker once.

Expected result / pass-fail rule: One mock audit receipt per action and stable references; no real delivery claim.

Actual status and evidence: G07 proves two approved-letter worker executions and dispatch variables on the selected route. The redirected-referrer branch, direct audit-row count and retry test are NOT EVIDENCED.

Clinical change, Finance and refund (BR-20/21/24/38)

Preconditions, data and actions: Final model; authorised money-affected change, Finance decision resolved, amount 25.50; inspect result and H2.

Expected result / pass-fail rule: D notification does not refund automatically; A refund follows Finance; one ledger/audit receipt and stable retry.

Actual status and evidence: A/D Java tests pass in isolation; final cross-member route NOT EVIDENCED.

4. Evaluation and release actions

The evidence now supports one complete live final-model self-pay route, selected historical Camunda routes and current A/D worker tests. This is not unconditional acceptance of every branch. Gaps remain material: the role-permission criterion failed in the earlier recorded test; B's pending-to-ready booking state logic requires correction and database tests; C's redirect/referrer and retry behaviour lacks a recorded final-model run; A's urgent/report/refund branches and D's enquiry/change branches need final-version route records. `docs/requirements-traceability-matrix.md` should be updated so supported, partially supported and unsupported requirements match these findings.

Before final group sign-off, the group should (1) fix and test B's pending-to-confirmed booking transition; (2) add browser Tasklist form observations and direct H2 row checks to the recorded G07 API run if those claims will be presented; (3) run the B/C owner-specific, role-boundary, A refund/urgent and D change/enquiry scenarios; (4) obtain second-owner review of documents and results; and (5) check any strategic/i* artefacts required by the definitive brief. The pack is already on `main`; G07 itself has already reached COMPLETED. Do not mark the remaining criteria Done merely because source files exist.